

AP Police SI Prelims 2026 — Practice Question Sets

Topic-wise MCQs for Paper 1 (Arithmetic & Reasoning) with full solutions. Try each question first, then check the worked solution that follows each set.

ARITHMETIC

Set A — Percentage

- 35% of 240 = ? (a) 78 (b) 84 (c) 90 (d) 96
- If 60 is 25% of a number, the number is: (a) 200 (b) 240 (c) 180 (d) 150
- A number increased by 25% becomes 250. The number is: (a) 180 (b) 200 (c) 210 (d) 220
- A salary rises 20% then falls 20%. Net change: (a) 0% (b) -4% (c) +4% (d) -2%
- In an exam, a student scores 180 out of 300. Percentage = (a) 55% (b) 60% (c) 62% (d) 65%
- If the price of rice rises 25%, by what % must consumption fall to keep spend the same? (a) 20% (b) 25% (c) 18% (d) 22%

Solutions A: 1. (b) $0.35 \times 240 = 84$. 2. (b) $N = 60/0.25 = 240$. 3. (b) $1.25N = 250 \rightarrow N = 200$. 4. (b) Net = $-(20^2/100) = -4\%$. 5. (b) $180/300 \times 100 = 60\%$. 6. (a) Reduction% = $25/(100+25) \times 100 = 20\%$.

Set B — Profit & Loss

- CP = 250, SP = 300. Profit% = (a) 18% (b) 20% (c) 22% (d) 25%
- An article sold at 480 gives 20% profit. CP = (a) 380 (b) 400 (c) 420 (d) 440
- Successive discounts of 20% and 10% equal a single discount of: (a) 28% (b) 30% (c) 25% (d) 32%
- A man sells at a 10% loss for 540. CP = (a) 580 (b) 600 (c) 620 (d) 560
- Buying 12 for 10 and selling 10 for 12, profit% = (a) 40% (b) 44% (c) 20% (d) 36%

Solutions B: 1. (b) $50/250 \times 100 = 20\%$. 2. (b) $CP = 480 \times 100/120 = 400$. 3. (a) $20+10-(20 \times 10)/100 = 28\%$. 4. (b) $CP = 540 \times 100/90 = 600$. 5. (b) CP per unit = $10/12$; SP per unit = $12/10$. Profit% = $(1.2-0.8333)/0.8333 \times 100 = 44\%$.

Set C — Simple & Compound Interest

- SI on 4000 at 5% for 3 yrs = (a) 540 (b) 600 (c) 620 (d) 660
- A sum doubles in 8 yrs at SI. Rate = (a) 10% (b) 12% (c) 12.5% (d) 15%
- CI on 1000 at 10% for 2 yrs = (a) 200 (b) 210 (c) 220 (d) 215
- For 2 yrs, CI - SI on P at 10% = 50. Then P = (a) 4000 (b) 5000 (c) 6000 (d) 4500
- A sum at 8% SI for 5 yrs gives 200 interest. Principal = (a) 400 (b) 500 (c) 600 (d) 450

Solutions C: 1. (b) $(4000 \times 5 \times 3)/100 = 600$. 2. (c) To double, $SI = P \rightarrow 100 = R \times 8 \rightarrow R = 12.5\%$. 3. (b) $1000(1.1)^2 - 1000 = 210$. 4. (b) $P(0.1)^2 = 50 \rightarrow 0.01P = 50 \rightarrow P = 5000$. 5. (b) $P = (200 \times 100)/(8 \times 5) = 500$.

Set D — Ratio, Proportion & Average

1. Divide 720 in ratio 5:4. Larger share = (a) 320 (b) 400 (c) 360 (d) 420
2. Average of 5 numbers is 18. Sum = (a) 80 (b) 90 (c) 95 (d) 100
3. Average of 10, 20, 30, 40 = (a) 22 (b) 25 (c) 28 (d) 30
4. $A:B = 2:3$, $B:C = 4:5$. Then $A:B:C =$ (a) 8:12:15 (b) 2:3:5 (c) 8:12:10 (d) 6:9:10
5. Average age of 4 members is 25; a 5th of age 35 joins. New average = (a) 26 (b) 27 (c) 28 (d) 30

Solutions D: 1. (b) $720 \times \frac{5}{9} = 400$. 2. (b) $18 \times 5 = 90$. 3. (b) $(10+20+30+40)/4 = 25$. 4. (a) Make B common: $A:B = 8:12$, $B:C = 12:15 \rightarrow 8:12:15$. 5. (b) $(100+35)/5 = 27$.

Set E — Time & Work / Speed & Distance

1. A does a job in 12 days, B in 6 days. Together = (a) 3 (b) 4 (c) 5 (d) 6 days
2. A train 150 m long at 20 m/s crosses a pole in: (a) 6.5 s (b) 7.5 s (c) 8 s (d) 7 s
3. 90 km/h in m/s = (a) 20 (b) 22 (c) 25 (d) 27
4. A covers 120 km in 3 hrs. Speed = (a) 35 (b) 40 (c) 45 (d) 50 km/h
5. A and B can finish in 10 and 15 days. Together they take: (a) 5 (b) 6 (c) 7 (d) 8 days

Solutions E: 1. (b) $1/12 + 1/6 = 3/12 = 1/4 \rightarrow 4$ days. 2. (b) $150/20 = 7.5$ s. 3. (c) $90 \times 5/18 = 25$ m/s. 4. (b) $120/3 = 40$ km/h. 5. (b) $1/10 + 1/15 = 1/6 \rightarrow 6$ days.

Set F — Mensuration & Number System

1. Area of a circle, $r = 7$ ($\pi = 22/7$): (a) 144 (b) 154 (c) 164 (d) 174
2. Volume of a cube of side 5 = (a) 100 (b) 115 (c) 125 (d) 150
3. HCF of 12 and 18 = (a) 3 (b) 6 (c) 9 (d) 12
4. LCM of 4 and 6 = (a) 10 (b) 12 (c) 18 (d) 24
5. Perimeter of a rectangle $8 \times 5 =$ (a) 24 (b) 26 (c) 28 (d) 40

Solutions F: 1. (b) $(22/7) \times 49 = 154$. 2. (c) $5^3 = 125$. 3. (b) HCF = 6. 4. (b) LCM = 12. 5. (b) $2(8+5) = 26$.

REASONING & MENTAL ABILITY

Set G — Series

1. 3, 6, 12, 24, ? (a) 36 (b) 48 (c) 30 (d) 40
2. 2, 5, 10, 17, 26, ? (a) 35 (b) 37 (c) 39 (d) 41
3. 1, 4, 9, 16, ? (a) 20 (b) 25 (c) 24 (d) 36
4. A, C, E, G, ? (a) H (b) I (c) J (d) K
5. 100, 96, 88, 76, ? (a) 60 (b) 64 (c) 58 (d) 62

Solutions G: 1. (b) $\times 2$ each $\rightarrow 48$. 2. (b) differences 3, 5, 7, 9 $\rightarrow +11 = 37$ (or n^2+1). 3. (b) squares $\rightarrow 25$. 4. (b) skip one letter $\rightarrow I$. 5. (a) differences $-4, -8, -12, -16 \rightarrow 76-16 = 60$.

Set H — Coding-Decoding

1. If CAT = 3-1-20, then DOG = (a) 4-15-7 (b) 4-14-7 (c) 3-15-7 (d) 4-15-8
2. If A=1...Z=26, SUN = (a) 19-21-14 (b) 18-21-14 (c) 19-20-14 (d) 19-21-13
3. If PEN is coded RGP (each letter +2), then BAT = (a) DCV (b) CCV (c) DCW (d) DBV

4. If 'RED' = 'GViD' style shift not given, simplest: reverse of 'DOG' = (a) GOD (b) DGO (c) ODG (d) GDO

Solutions H: 1. (a) D=4, O=15, G=7. 2. (a) S=19, U=21, N=14. 3. (a) B→D, A→C, T→V → DCV.
4. (a) Reverse of DOG = GOD.

Set I — Blood Relations & Direction

1. Pointing to a man, a woman says "He is the son of my mother's only son." The man is her:
(a) brother (b) son (c) nephew (d) cousin
2. A walks 3 km North then 4 km East. Distance from start: (a) 5 km (b) 6 km (c) 7 km (d) 4 km
3. If facing North you turn right twice, you now face: (a) East (b) West (c) South (d) North
4. B is A's father's sister's son. B is A's: (a) cousin (b) uncle (c) brother (d) nephew

Solutions I: 1. (c) Her mother's only son is her brother; his son is her nephew. 2. (a) $\sqrt{3^2+4^2} = 5$. 3. (c) Right twice from North = South. 4. (a) Father's sister's son = cousin.

Set J — Analogy & Odd One Out

1. Hand : Glove :: Foot : ? (a) Sock (b) Shoe (c) Leg (d) Toe
2. Doctor : Hospital :: Teacher : ? (a) Class (b) School (c) Student (d) Book
3. Odd one out: (a) Rose (b) Lotus (c) Mango (d) Lily
4. Pen : Write :: Knife : ? (a) Sharp (b) Cut (c) Kitchen (d) Blade

Solutions J: 1. (b) worn on → Shoe. 2. (b) workplace → School. 3. (c) Mango is a fruit; rest are flowers. 4. (b) used to → Cut.

How to Use These Sets

- Time yourself: aim for under 45 seconds per Arithmetic question and 30 seconds per Reasoning question.
- Re-attempt every question you got wrong after a week.
- This is **Practice Set 1** — ask for more sets per topic (with harder levels) any time.

Cross-check the exam pattern and difficulty against the official APSLPB notification.